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Multiple Studies: Aggregating support (= evidence synthesis)

References:

- Kuiper, R.M., Buskens, V.W., Raub, W., and Hoijtink, H. (2013). Combining statistical evidence from several studies: A method using Bayesian updating and an example from research on trust problems in social and economic exchange. *Sociological Methods and Research*, 42 (1), 60–81.
- Van Lissa, C. J., Clapper, E.-B., and Kuiper, R.M. (2024). Aggregating evidence from conceptual replication studies using the product Bayes factor. Pre-print: [10.31234/osf.io/nvqpw](https://doi.org/10.31234/osf.io/nvqpw)
- Kuiper, R.M. and Clapper, E.-B. (2025). Aggregating Evidence for a Central Theory From Diverse Studies Using the Generalized Order-Restricted Information Criterion. *Psychological Methods*. Pre-print: https://osf.io/preprints/psyarxiv/qv76x_v2

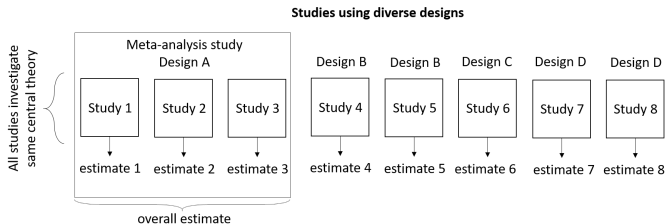
Motivation

In science, the gold standard for evidence is an empirical result that is consistent across multiple studies.

- **Replicability/Replication crisis** in social science.
- Political scientists call for meta-scientific introspection.

Therefore, need for aggregating results.

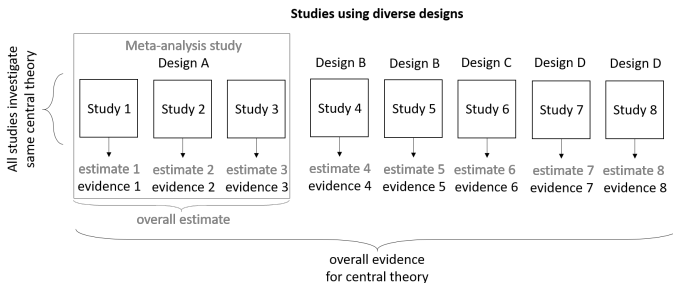
Current best practice



Current best practice is meta-analysis and Bayesian updating.

- Not applicable for diverse research designs.
- Not applicable for incomparable estimates.

Need for new methodology: Evidence Synthesis



Note: All studies do investigate the **same, central theory** (using diverse designs).

Trust Example: Meta-Analysis versus Evidence Synthesis

Study	Type of statistical model
1	univariate regression
2	univariate regression
3	probit regression
4	three-level logistic regression

Same design? e.g., same set of predictors?

Conceptual replications (studying **same, central theory**)!

	Meta-Analysis	Evidence Synthesis
Effect size not required		✓
Deal with diverse designs		✓
Main results (goal)	Estimate of effect size	Evidence for hypotheses
Check:		Same theoretical relationships?

Reference:

Kuiper, R.M., Buskens, V.W., Raub, W., and Hoijtink, H. (2013). Combining statistical evidence from several studies: A method using Bayesian updating and an example from research on trust problems in social and economic exchange. *Sociological Methods and Research*, 42 (1), 60–81.

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Example: 4 studies regarding one concept

Study	Type of study	Number of observations n	Type of model
1	survey	895 transactions	univariate regression
2	experiment	348 decisions by 40 subjects	univariate regression
3	experiment	1249 decisions by 125 subjects	probit regression
4	experiment	2160 decisions by 144 subjects	three-level logistic regression
Study	Outcome y (trust)		scale y
1	effort invested in management		ratio
2	effort invested in management		ratio
3	choice of vignettes		dummy
4	trustfulness		dummy
Study	Predictor x_1 (past / previous experience)		scale x_1
1	existence relationship with supplier		dummy
2	type of relationship with supplier		interval
3	bought a car from The Autoshop before		dummy
4	number of times a trustee honored trust in the past		ratio
Study	some of the other predictors		
1	transaction characteristics, expected future transactions, network embeddedness		
2	transaction characteristics, expected future transactions, network embeddedness		
3	expected future transactions, network embeddedness		
4	future interactions, network embeddedness		

One-Parameter Example: Hypotheses of interest

Main central theory

Previous experience has a positive effect on trust.

For simplicity, only one relationship here, could have been more.

Study-specific hypothesis

$$\beta_1 > 0$$

Here, for each study the same hypothesis.

Set of central theories

H_0 : no effect,

$H_{>}$: positive effect,

$H_{<}$: negative effect.

Note 1: Central hypotheses for all studies, not w.r.t. average parameter.

Note 2: In practice, I would not include H_0 ...

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Example: Trust (y) & previous experience (x_1)

Possible input GORIC(A) evidence synthesis
(using *evSyn* function in R package *restriktor*):

- Parameter estimates and their covariance matrix

t	$\hat{\beta}_1$	$\hat{\sigma}_{\beta_1}$
1	0.090	0.029
2	0.140	0.054
3	1.090	0.093
4	1.781	0.179

Note: Here, one parameter (β_1); thus, cov. matrix $\hat{\beta}_1 = \text{variance } \hat{\beta}_1 = \hat{\sigma}_{\beta_1}^2$ (not $\hat{\sigma}_{\beta_1}$)

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One-Parameter Example: Results per study

using GORICA

Results **per study** (so, not aggregated yet)!

Study-specific support for **central theory**:

Table: GORICA weights ($w_{t,m}$) for Hypothesis H_m in Study t

m / t	$w_{t,m}$			
	1	2	3	4
0	0.013	0.052	0.000	0.000
>	0.979	0.916	1.000	1.000
<	0.008	0.032	0.000	0.000

Note: Weight is at max 1.

So, here, conclusion beforehand clear.... but no quantification yet.

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- $w_{4,>}^1 = 1 \Rightarrow$ full support for $H_>$
 $w_{4,0}^1 = w_{4,<}^1 = 0 \Rightarrow$ no support for H_0 and $H_<$
- Support for $H_>$ ($w_{4,>}^1$) is highest: favor $H_>$ over H_0 and $H_<$.
- Hence, previous experience has a positive effect on trust, which receives full support.

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Example R code

Input: parameter estimates

```
# Simple example:
#- 4 studies;
#- in all studies, 2 parameters of interest;
#- in all studies, 2 parameters labelled as beta1 and beta2:
#
# Study 1
est_1 <- c(beta1 = 0.09, beta2 = 0.16)
# or:
#est_1 <- c(0.09, 0.16); names(est_1) <- c("beta1", "beta2")
# Study 2
est_2 <- c(beta1 = 0.14, ...)
# other studies
...
#
# List of estimates for all 4 studies:
Param_studies <- list(est_1, est_2, est_3, est_4)
```

Example R code

Input: covariance matrices of parameter estimates

```
# Simple example:
#- 4 studies;
#- in all studies, 2 parameters of interest;
#- in all studies, 2 parameters labelled as beta1 and beta2:

# Covariance matrix of the 2 parameter estimates
# for each of the 4 studies:
vcov_est_1 <- matrix(c(0.029^2, 0.005, 0.005, 0.036^2),
                     nrow = 2)
vcov_est_2 <- matrix(c(0.054^2, ...), nrow = 2)
...
#
# list of covariance matrices:
CovMx_studies <- list(vcov_est_1, vcov_est_2,
                      vcov_est_3, vcov_est_4)
```

Example R code

Input: (study-specific) hypotheses

```
# For ease, all studies have the same hypotheses in the set
# (here, consisting of the following two hypotheses):
H_bigger <- "beta2 > beta1"
H_smaller <- "beta2 < beta1"
#
# Here, the hypotheses are the same for all studies,
# then, only specify one list:
Hypo_studies <- list(H_bigger = H_bigger,
                     H_smaller = H_smaller)
# Note: see `Tutorial_GORIC_restriktor_evSyn.html`
# for specifying study-specific hypotheses.

# Here, the 2 hypotheses cover the whole space of theories.
# Therefore, we do not need a safeguard-hypothesis here:
safeguard <- "none"
# Note: We could have used H_bigger with its complement.
```

Example R code

GORICA evidence synthesis

```
# GORICA evidence synthesis #  
  
#library(restriktor)  
evSyn_trust <- evSyn(object = Param_studies,  
                     VCOV = CovMx_studies,  
                     hypotheses = Hypo_studies,  
                     #type = "added", # Default  
                     comparison = safeguard)  
  
evSyn_trust  
#summary(evSyn_trust)  
plot(evSyn_trust)
```

More types of input and other options can be found in
'Tutorial_GORIC_restriktor_evSyn.html' available from
<https://github.com/rebeccakuiper/Tutorials>.

Hands-on/Demo (1): Evidence Synthesis on trust example

Let's practice (in Rstudio):

- If needed: <https://github.com/rebeccakuiper/Tutorials>:
 1. Click on green button called Code.
 2. Download zip (last option in list).
 3. Unzip it on your machine (that folder is now your working dir.).
- Open 'Tutorial_GORIC_restriktor_evSyn.html' and 'Hands-on_5_GORIC_evSyn_restriktor.R' (in 'Hands-on files').
- Install packages and load them.
- In first example:
 - Read and inspect data.
 - Specify (study-specific) hypothesis/-es, which thus may differ per study, but all reflect same central theory.
- Aggregate evidence using `evSyn()`.
- Inspect and interpret output.



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Two-Parameter Example

(example metafor and thus homogeneous designs)

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Example metafor: Berkey et al. (1998)

Surgical and non-surgical treatments for medium-severity periodontal disease are compared in five trials (i.e., 5 studies).

Two outcomes reflect severity:

- attachment level (AL) and
- probing depth (PD)

one year after the treatment.

Higher attachment levels and lower probing depth denote less severity.

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Two-Parameter Example ctd.

Example metafor: Berkey et al. (1998)

Effect sizes (θ):

(raw) mean differences (ref. cat. is non-surgical treatment).

Positive value (θ_{AL} & θ_{PD}):

surgery more effective than non-surgical treatment (for AL and PD).

Surgery more effective in reducing severity ($\theta_{AL} > 0$ & $\theta_{PD} > 0$):

increasing the attachment level (AL) and

decreasing the probing depth (DP).

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Two-Parameter Example: Possible sets

Possible expectations and hypotheses (central theories):

- Set 1 Based on sign of effect sizes:
Non-surgery more effective for attachment level ($\theta_{AL} < 0$)
Surgery more effective for probing depth ($\theta_{PD} > 0$).
 $H_{-+} : \theta_{AL} < 0; \theta_{PD} > 0$
Possibly a competing hypothesis.
- Set 2 Use magnitude of the effect size; e.g.,:
 $H_{minAbsES} : |\theta_{AL}| > 0.2; |\theta_{PD}| > 0.2$.
Possibly a competing hypothesis.
- Set 3 Comparison of absolute sizes; e.g.,:
 $H_{absComp} : |\theta_{AL}| > |\theta_{PD}|$.

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Two-Parameter Example: Hypothesis (central theory)

Let us assume that we are interested in the third set of hypotheses (central theories):

$$H_{absComp} : |\theta_{AL}| > |\theta_{PD}|.$$

Let us further assume that there are no competing hypotheses. Thus, $H_{absComp}$ will be evaluated against its complement:

$$H_c : \text{not } H_{absComp}.$$

$$\text{Here, } H_c : |\theta_{AL}| < |\theta_{PD}|.$$

Next, also use notation $H_{>}$ & $H_{<}$, resp.

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Results **per study** (so, not aggregated yet)!

Study-specific support for **central theory**:

Table: GORICA weights ($w_{t,m}$) for Hypothesis H_m in Study t

m / t	$w_{t,m}$				
	1	2	3	4	5
$>$	0.370	1.000	0.000	0.550	0.440
$<$	0.630	0.032	1.000	0.450	0.560

Note: Weight is at max 1.

So, here, mixed results.

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	$w_{t,m}^1$				
m / t	1	2	3	4	5
$>$	0.370	1.000	0.752	0.787	0.744
$<$	0.630	0.000	0.248	0.213	0.256

Note: Weight is at max 1.

- Support for $H_{>}$ ($w_{5,>}^1$) is highest: favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
- $H_{absComp}$ is $0.744/0.256 \approx 2.9$ times more likely than its complement.
- Preference central theory 'the difference in symptom reduction between surgery and non-surgery is larger for AI than for PB'

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$>$	0.370	1.000	0.752	0.787	0.744
$<$	0.630	0.000	0.248	0.213	0.256

Note: Weight is at max 1.

- Support for $H_>$ ($w_{5,>}^1$) is highest: favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
- $H_{absComp}$ is $0.744/0.256 \approx 2.9$ times more likely than its complement.
- Preference central theory 'the difference in symptom reduction between surgery and non-surgery is larger for AI than for PB'

Two-Parameter Example: Results & Conclusions

Aggregated support for **central theory**:

Table: Overall GORICA weights ($w_{t,m}^1$) for Hypothesis H_m in Study t

m / t	$w_{t,m}^1$				
	1	2	3	4	5
$>$	0.370	1.000	0.752	0.787	0.744
$<$	0.630	0.000	0.248	0.213	0.256

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Note: Weight is at max 1.

- Support for $H_{>}$ ($w_{5,>}^1$) is highest:
favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
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Two-Parameter Example: Results & Conclusions

Aggregated support for **central theory**:

Table: Overall GORICA weights ($w_{t,m}^1$) for Hypothesis H_m in Study t

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Note: Weight is at max 1.

- Support for $H_{>}$ ($w_{5,>}^1$) is highest:
favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
- $H_{absComp}$ is $0.744/0.256 \approx 2.9$ times more likely than its complement.
- Preference central theory 'the difference in symptom reduction between surgery and non-surgery is larger for AL than for PB'.

Two-Parameter Example: Results & Conclusions

Aggregated support for **central theory**:

Table: Overall GORICA weights ($w_{t,m}^1$) for Hypothesis H_m in Study t

m / t	$w_{t,m}^1$				
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Note: Weight is at max 1.

- Support for $H_{>}$ ($w_{5,>}^1$) is highest:
favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
- $H_{absComp}$ is $0.744/0.256 \approx 2.9$ times more likely than its complement.
- Preference central theory 'the difference in symptom reduction between surgery and non-surgery is larger for AL than for PB'.

Two-Parameter Example: Results & Conclusions

Aggregated support for **central theory**:

Table: Overall GORICA weights ($w_{t,m}^1$) for Hypothesis H_m in Study t

m / t	$w_{t,m}^1$				
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$>$	0.370	1.000	0.752	0.787	0.744
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Note: Weight is at max 1.

- Support for $H_{>}$ ($w_{5,>}^1$) is highest:
favor $H_{absComp}$ over its complement.
- On study level: mixed support; combined: preference for $H_{absComp}$.
- $H_{absComp}$ is $0.744/0.256 \approx 2.9$ times more likely than its complement.
- Preference central theory 'the difference in symptom reduction between surgery and non-surgery is larger for AL than for PB'.

Example R code

Input: data.frame

Instead of lists with parameter estimates and covariance matrices, can also use a data.frame.

```
# data.frame with info on all 5 studies (trials)
#
#if (!require("metafor")) install.packages("metafor")
#library(metafor)
dat <- dat.berkey1998
```

Example R code

Input: (study-specific) hypotheses

Parameter names to be used in the hypotheses:

- If one outcome: 'theta'.
- Labels used in specified outcome column/variable.
Here, 'AL' and 'PD', levels of the variable 'outcome'.

```
# All studies have the same hypotheses in the set:  
H_absComp_Gr <- "abs(AL) > abs(PD)"  
# Complement of H_absComp_Gr, included by default.  
#  
# Here, the hypotheses are the same for all studies,  
# then, only specify one list:  
Hypo_studies <- list(H_absComp = H_absComp_Gr)  
# Note: see `Tutorial_GORIC_restriktor_evSyn.html`  
# for specifying study-specific hypotheses.
```

Example R code

GORICA evidence synthesis

```
# GORICA evidence synthesis #  
  
#library(restriktor)  
results_Set3_Gr <- evSyn(object = dat,  
  outcome_col = "outcome", # column with 'AL' & 'PB'  
  hypotheses = Hypo_studies)  
#type = "added", # Default  
#comparison = "complement" # Default  
  
#results_Set3_Gr  
summary(results_Set3_Gr)  
plot(results_Set3_Gr)
```

More types of input and other options can be found in
'Tutorial.GORIC_restriktor_evSyn.html' available from
<https://github.com/rebeccakuiper/Tutorials>.

Hands-on/Demo (2): Evidence Synthesis on Berkey example (data.frame)

Let's practice (in Rstudio):

- If needed: <https://github.com/rebeccakuiper/Tutorials>.
- Open 'Tutorial_GORIC_restriktor_evSyn.html' and 'Hands-on_5_GORIC_evSyn_restriktor.R' (in 'Hands-on files').
- Install packages and load them.
- Check-out Berkey example:
 - Read and inspect data (dat.berkey1998).
 - Specify (study-specific) hypothesis/-es (make up your own), which thus may differ per study, but all reflect same central theory.
Note: Check which labels are needed for the parameters.
- Aggregate evidence using evSyn().
- Inspect and interpret output.

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Evidence synthesis

Two-Parameter Example: Example metafor

evSyn options

Evidence synthesis approaches: Added and Equal

End

Updating hypotheses & Evidence synthesis

More...

evSyn option: (re-)order studies

evSyn plot for original study order

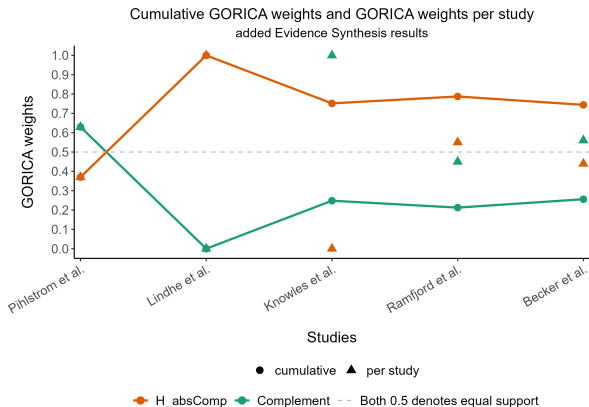


Figure: evSyn plot for original study order.

evSyn option: (re-)order studies

order_studies argument in *evSyn*

One can re-order studies using the *order_studies* argument in *evSyn*:

- Pre-specified: *order_studies* = "ascending" or "descending".
- Or based on own order: *order_studies* = *<insert variable name>*; e.g.:

```
order_ascYear <- order(dat[dat$outcome=="AL",]$year)
```

```
results_Set3_Gr_ascYear <- evSyn(object = dat,  
                                outcome_col = "outcome",  
                                hypotheses = Hypo_studies,  
                                order_studies = order_ascYear)
```

```
#results_Set3_Gr_ascYear  
summary(results_Set3_Gr_ascYear)  
plot(results_Set3_Gr_ascYear)
```

Re-ordering of course does not affect the overall evidence/support.

evSyn option: (re-)order studies

order_studies argument in *evSyn*

One can re-order studies using the *order_studies* argument in *evSyn*:

- Pre-specified: *order_studies* = "ascending" or "descending".
- Or based on own order: *order_studies* = *<insert variable name>*; e.g.:

```
order_ascYear <- order(dat[dat$outcome=="AL",]$year)
```

```
results_Set3_Gr_ascYear <- evSyn(object = dat,  
                                outcome_col = "outcome",  
                                hypotheses = Hypo_studies,  
                                order_studies = order_ascYear)
```

```
#results_Set3_Gr_ascYear  
summary(results_Set3_Gr_ascYear)  
plot(results_Set3_Gr_ascYear)
```

Re-ordering of course does not affect the overall evidence/support.

evSyn option: (re-)order studies

order_studies argument in *evSyn*

One can re-order studies using the *order_studies* argument in *evSyn*:

- Pre-specified: *order_studies* = "ascending" or "descending".
- Or based on own order: *order_studies* = <insert variable name>; e.g.:

```
order_ascYear <- order(dat[dat$outcome=="AL",]$year)
```

```
results_Set3_Gr_ascYear <- evSyn(object = dat,  
  outcome_col = "outcome",  
  hypotheses = Hypo_studies,  
  order_studies = order_ascYear)
```

```
#results_Set3_Gr_ascYear  
summary(results_Set3_Gr_ascYear)  
plot(results_Set3_Gr_ascYear)
```

Re-ordering of course does not affect the overall evidence/support.

evSyn option: (re-)order studies

order_studies argument in *evSyn*

One can re-order studies using the *order_studies* argument in *evSyn*:

- Pre-specified: *order_studies* = "ascending" or "descending".
- Or based on own order: *order_studies* = <insert variable name>; e.g.:

```
order_ascYear <- order(dat[dat$outcome=="AL",]$year)

results_Set3_Gr_ascYear <- evSyn(object = dat,
                                outcome_col = "outcome",
                                hypotheses = Hypo_studies,
                                order_studies = order_ascYear)

#results_Set3_Gr_ascYear
summary(results_Set3_Gr_ascYear)
plot(results_Set3_Gr_ascYear)
```

Re-ordering of course does not affect the overall evidence/support.

evSyn option: (re-)order studies

order_studies argument in *evSyn*

One can re-order studies using the *order_studies* argument in *evSyn*:

- Pre-specified: *order_studies* = "ascending" or "descending".
- Or based on own order: *order_studies* = <insert variable name>; e.g.:

```
order_ascYear <- order(dat[dat$outcome=="AL",]$year)

results_Set3_Gr_ascYear <- evSyn(object = dat,
                                outcome_col = "outcome",
                                hypotheses = Hypo_studies,
                                order_studies = order_ascYear)

#results_Set3_Gr_ascYear
summary(results_Set3_Gr_ascYear)
plot(results_Set3_Gr_ascYear)
```

Re-ordering of course does not affect the overall evidence/support.

evSyn option: (re-)order studies

evSyn plot for re-ordered studies

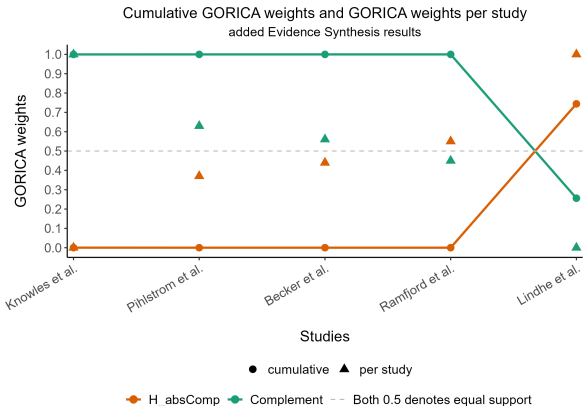


Figure: evSyn plot for studies ordered on ascending support overall best hypothesis.

evSyn option: leave 1 study out

leave1studyout

leave1studyout: insight into possible outlier or influential studies;
a sensitivity analyses on the GORICA Evidence Synthesis.

```
leave1out_Set3_Gr <- leave1studyout(results_Set3_Gr)  
#leave1out_Set3_Gr  
leave1out_Set3_Gr$OverallGoricaWeights
```

evSyn option: leave 1 study out

leave1studyout

leave1studyout: insight into possible outlier or influential studies;
a sensitivity analyses on the GORICA Evidence Synthesis.

```
leave1out_Set3_Gr <- leave1studyout(results_Set3_Gr)  
#leave1out_Set3_Gr  
leave1out_Set3_Gr$OverallGoricaWeights
```

evSyn option: leave 1 study out

(part of) output *leave1studyout*

```
$OverallGoricaWeights
```

	H_absComp	Complement
Leave study nr. 1 out:	0.831699126	0.1683008737
Leave study nr. 2 out:	0.000189393	0.9998106070
Leave study nr. 3 out:	0.999884608	0.0001153924
Leave study nr. 4 out:	0.703796842	0.2962031578
Leave study nr. 5 out:	0.787430604	0.2125693956

```
$OverallPrefHypo
```

```
Leave study nr. 1 out: "H_absComp"  
Leave study nr. 2 out: "Complement"  
...
```

Here, results sensitive to the choice of set of studies.
Conclusion even changes when Study 2 is left out.

evSyn option: leave 1 study out

(part of) output *leave1studyout*

```
$OverallGoricaWeights
```

	H_absComp	Complement
Leave study nr. 1 out:	0.831699126	0.1683008737
Leave study nr. 2 out:	0.000189393	0.9998106070
Leave study nr. 3 out:	0.999884608	0.0001153924
Leave study nr. 4 out:	0.703796842	0.2962031578
Leave study nr. 5 out:	0.787430604	0.2125693956

```
$OverallPrefHypo
```

```
Leave study nr. 1 out: "H_absComp"  
Leave study nr. 2 out: "Complement"  
...
```

Here, results sensitive to the choice of set of studies.

Conclusion even changes when Study 2 is left out.

evSyn option: leave 1 study out

(part of) output *leave1studyout*

```
$OverallGoricaWeights
```

	H_absComp	Complement
Leave study nr. 1 out:	0.831699126	0.1683008737
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```
$OverallPrefHypo
```

```
Leave study nr. 1 out: "H_absComp"  
Leave study nr. 2 out: "Complement"  
...
```

Here, results sensitive to the choice of set of studies.
Conclusion even changes when Study 2 is left out.

Hands-on/Demo (3): Evidence Synthesis ('evSyn') options

Let's practice (in Rstudio):

- If needed: <https://github.com/rebeccakuiper/Tutorials>.
- Open 'Tutorial_GORIC_restriktor_evSyn.html' and 'Hands-on_5_GORIC_evSyn_restriktor.R' (in 'Hands-on files').
- Install packages and load them.
- Check-out Berkey example:
 - Read and inspect data (dat.berkey1998).
 - Specify (study-specific) hypothesis/-es (make up your own), which thus may differ per study, but all reflect same central theory.
Note: Check which labels are needed for the parameters.
- Aggregate evidence using evSyn().
- **Play around with evSyn() options.**
- Inspect and interpret output.

Two approaches: Added- vs Equal-evidence approach

Situation A: Evidence from 5 studies with $n = 100$.

Situation B: Evidence from 1 study with $n = 500$.

Approach 1: Situation A is stronger/extremer than Situation B

Conclusion: Evidence theory true in all studies.

Then, as we did before: Added-evidence approach.

Approach 2: Situation A is equally strong as Situation B

(cf. meta-analysis)

Conclusion: Evidence theory true on average.

Then, alternative method needed: Equal-evidence approach.

Two approaches: Added- vs Equal-evidence approach

Situation A: Evidence from 5 studies with $n = 100$.

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Two approaches: Added- vs Equal-evidence approach

Situation A: Evidence from 5 studies with $n = 100$.

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Approach 2: Situation A is equally strong as Situation B

(cf. meta-analysis)

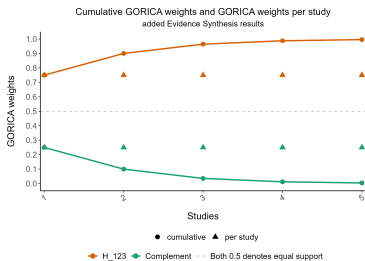
Conclusion: Evidence theory true on average.

Then, alternative method needed: Equal-evidence approach.

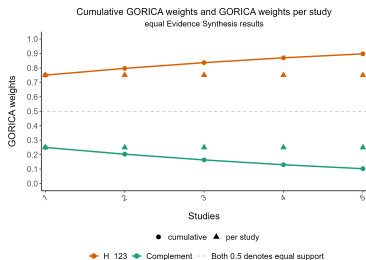
Added- vs Equal-evidence approach

5 studies, each $n = 100$.

Added



Equal



Regression model with 3 predictors ($\rho = .1$) and an intercept;
 $f^2 = 0.05$ (small) and thus $R^2 \approx 0.05$;
 $H_{123} : \beta_1 < \beta_2 < \beta_3$ versus its complement.

Example R code

GORICA equal-evidence synthesis

```
# GORICA equal-evidence synthesis #  
  
#library(restriktor)  
results_Set3_Gr_equal <- evSyn(object = dat,  
  outcome_col = "outcome", # column with 'AL' & 'PB'  
  hypotheses = Hypo_studies,  
  type_ev = "equal") # Now, equal-evidence approach  
  
#results_Set3_Gr_equal  
summary(results_Set3_Gr_equal)  
plot(results_Set3_Gr_equal)
```

More types of input and other options can be found in
'Tutorial_GORIC_restriktor_evSyn.html' available from
<https://github.com/rebeccakuiper/Tutorials>.

Hands-on/Demo (4): Equal-evidence Synthesis on Berkey example (data.frame)

Let's practice (in Rstudio):

- If needed: <https://github.com/rebeccakuiper/Tutorials>.
- Open 'Tutorial_GORIC_restriktor_evSyn.html' and 'Hands-on_5_GORIC_evSyn_restriktor.R' (in 'Hands-on files').
- Install packages and load them.
- In some of the examples:
 - Read and inspect data.
 - Specify (study-specific) hypothesis/-es (make up your own), which thus may differ per study, but all reflect same central theory.
Note: Check which labels are needed for the parameters.
- Aggregate evidence using `evSyn()`.
Now, use 'type_ev = "equal"'.
- Inspect and interpret output.

The End

Multiple studies

Thanks for listening!

Are there any questions?

Websites

<https://github.com/rebeccakuiper/Tutorials>

www.uu.nl/staff/RMKuiper/Extra1

www.uu.nl/staff/RMKuiper/Extra2

informative-hypotheses.sites.uu.nl/software/goric/

E-mail

r.m.kuiper@uu.nl

What's next

Depending on time and wishes:

- Demo in R
- Inspect 'Tutorial_GORIC_restriktor_evSyn.html' from <https://github.com/rebeccakuiper/Tutorials>

We end with:

- Lab

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Updating hypotheses & Evidence synthesis

More...

Example

Zondervan-Zwijnenburg et al. (2020)

Example based on results in Zondervan-Zwijnenburg et al. (2020):

RQ: Can age of the mother predict externalizing problem behavior of children around the age of 11.

(rated by the mother using the CBCL child behavior checklist)

Studied by 3 cohort studies in the Netherlands:

TRAILS (N=1955), NTR (N=21921), and GEN-R (N=4549).

Reference:

Zondervan-Zwijnenburg et al. (2020). Parental Age and Offspring Childhood Mental Health: A Multi-Cohort, Population-Based Investigation. *Child Development*. 91(3), 964-982.

Example: Notes

Goal

Each of the cohorts measured the variables in their own way:
so, different operationalisation of same constructs.

Hence, cannot use meta-analysis nor Bayesian updating.

They did not want evidence for pattern on average,
but evidence that pattern exist in each of the three studies.

Updating hypotheses & Evidence synthesis

4 steps

Steps:

1. Randomly divide the data of each cohort into an exploratory and confirmatory part.
2. Use the exploratory data to construct informative hypotheses.
3. Use the confirmatory data to evaluate the informative hypotheses.
4. Evidence synthesis: Combine the results obtained for the three cohorts into one overall conclusion.

Updating hypotheses & Evidence synthesis: Example

Step 1

After randomly choosing 50% of each data set (the exploration set), the following results were obtained for each cohort:

Cohort	$\hat{\beta}_1$	p-val	$\hat{\beta}_2$	p-val	R^2
Gen-R	-.10	<.001	.02	<.001	.02
NTR	-.11	<.001	.06	<.001	.02
TRAILS	-.13	<.001	.06	.06	.02

The model was:

$$\text{CBCL} = \beta_0 + \beta_1 \text{age} + \beta_2 \text{age}^2 + \text{error} \quad (1)$$

Updating hypotheses & Evidence synthesis: Example

Step 1

Cohort	$\hat{\beta}_1$	p-val	$\hat{\beta}_2$	p-val	R^2
Gen-R	-.10	<.001	.02	<.001	.02
NTR	-.11	<.001	.06	<.001	.02
TRAILS	-.13	<.001	.06	.06	.02

Updated hypothesis:

- Significance and sign imply: $\beta_1 < 0$ & $\beta_2 > 0$.

Competing hypotheses (optional):

- Because effects seem small: $\beta_1 = 0$ & $\beta_2 = 0$.
- Because $\hat{\beta}_2$ not always significant: $\beta_1 < 0$ & $\beta_2 = 0$.

Updating hypotheses & Evidence synthesis: Example

Step 1

Cohort	$\hat{\beta}_1$	p-val	$\hat{\beta}_2$	p-val	R^2
Gen-R	-.10	<.001	.02	<.001	.02
NTR	-.11	<.001	.06	<.001	.02
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Updated hypothesis:

- Significance and sign imply: $\beta_1 < 0$ & $\beta_2 > 0$.

Competing hypotheses (optional):

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Updating hypotheses & Evidence synthesis: Example

Step 1

Cohort	$\hat{\beta}_1$	p-val	$\hat{\beta}_2$	p-val	R^2
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NTR	-.11	<.001	.06	<.001	.02
TRAILS	-.13	<.001	.06	.06	.02

Updated hypothesis:

- Significance and sign imply: $\beta_1 < 0$ & $\beta_2 > 0$.

Competing hypotheses (optional):

- Because effects seem small: $\beta_1 = 0$ & $\beta_2 = 0$.
- Because $\hat{\beta}_2$ not always significant: $\beta_1 < 0$ & $\beta_2 = 0$.

Updating hypotheses & Evidence synthesis: Example

Step 1

Cohort	$\hat{\beta}_1$	p-val	$\hat{\beta}_2$	p-val	R^2
Gen-R	-.10	<.001	.02	<.001	.02
NTR	-.11	<.001	.06	<.001	.02
TRAILS	-.13	<.001	.06	.06	.02

Updated hypothesis:

- Significance and sign imply: $\beta_1 < 0$ & $\beta_2 > 0$.

Competing hypotheses (optional):

- Because effects seem small: $\beta_1 = 0$ & $\beta_2 = 0$.
- Because $\hat{\beta}_2$ not always significant: $\beta_1 < 0$ & $\beta_2 = 0$.

Updating hypotheses & Evidence synthesis: Example

Step 1

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Updating hypotheses & Evidence synthesis: Example

Step 2

Set of competing informative hypotheses:

$$H_3 : \beta_1 < 0 \ \& \ \beta_2 > 0,$$

that is, the older the mothers the less externalizing problems occur, and, the rate of decrease 'decreases' with age;

$$H_1 : \beta_1 = 0 \ \& \ \beta_2 = 0,$$

that is, age cannot be used to predict externalizing problems;

$$H_2 : \beta_1 < 0 \ \& \ \beta_2 = 0,$$

that is, there is only a linear effect of age; and,

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Updating hypotheses & Evidence synthesis: Example

Step 3 - using GORICA

For each of H_1 , H_2 , H_3 , and H_u ,
the GORICA weights are computed (denoted w_m for H_m).

Updating hypotheses & Evidence synthesis: Example

Steps 3 and 4 - using GORICA

Using the confirmation set,
that is, the second 50% of the data of each of the three cohorts,
the following GORICA weights were obtained:

Cohort	w_1	w_2	w_3	w_u
Gen-R	.82	.04	.10	.05
NTR	.00	.97	.02	.01
TRAILS	.00	.88	.09	.03
All	.00	.99	.01	.00

Updating hypotheses & Evidence synthesis: Example

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Conclusion:

Based on the combined evidence in the three cohorts, there is almost full support for $H_2 : \beta_1 < 0 \text{ \& } \beta_2 = 0$.

That is, there is only a linear effect of age of the mother on externalizing problem behavior of children around the age of 11.

Updating hypotheses & Evidence synthesis: Example

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More...

More...

Possible types of sets of studies

- Conceptual replications of same authors, done as a robustness check (cf. triangulation).
- Searching for direct and indirect/conceptual replications in the literature.
- Using multiple $N = 1$ studies.
- Using different cohorts, where one can measure the variables in their own way.
- Using different subpopulations, possibly using different operationalisations.
- ...

Possible type of sets of studies (1/5)

- Conceptual replications of same authors, done as a robustness check (cf. triangulation).

This was done in the Trust example of Buskens and Raub.

Note: There, the central hypotheses regard one parameter of interest (in each study), but one can compare (absolute values of) multiple parameters or multiple effect sizes (Which may also differ per study).

For some examples, download 'Tutorial_GORIC_restriktor_evSyn.html' from <https://github.com/rebeccakuiper/Tutorials>.

Note: On github site, go to Code (green button) and download zip.

Possible type of sets of studies (2/5)

- Searching for direct and indirect/conceptual replications in the literature.

E.g., using the central hypothesis that the absolute strength ($|.$) of the relationship of communication competence (C) with willingness to communicate in a second language (WTC) is greater than the absolute strength of the relation of communication anxiety (A) with WTC, which is greater than the absolute strength of the relation of motivation (M) with WTC, that is, $|C| > |A| > |M|$; where C, A, and M are operationalized differently in the studies.

— 'Article': Example in bachelor thesis of Martijn Sips

— R scripts: <https://github.com/rebeccakuiper/Tutorials/tree/main/Examples%20evSyn/Example%20WtC>

Note: On github site, go to Code (green button) and download zip.

Possible type of sets of studies (3/5)

- Using multiple $N = 1$ studies.

– Article: Klaassen et al. (2018).

All for one or some for all? Evaluating informative hypotheses using multiple $N = 1$ studies. *Behavior Research Methods*, 50, 2276–2291.

<https://link.springer.com/article/10.3758/s13428-017-0992-5>.

Possible type of sets of studies (4/5)

- Using different cohorts, where one can measure the variables in their own way.

So, possibly using different operationalisation of the same constructs.

— Article: Zondervan-Zwijnenburg et al. (2020).

Parental Age and Offspring Childhood Mental Health: A Multi-Cohort, Population-Based Investigation. *Child Development*. 91(3), 964–982.

Possible type of sets of studies (5/5)

- Using different subpopulations, possibly using different operationalisations.

E.g., the Municipal Health Services (Dutch acronym: GGD) studied the positive consequences of corona on

- loneliness (a),
- mental health (b), and
- stress (c);

conditional on sex, age, and health.

Central hypothesis: $a < b < c$.

— R scripts: <https://github.com/rebeccakuiper/Tutorials/tree/main/Examples%20evSyn/Example%20corona%20GGD>

Note: On github site, go to Code (green button) and download zip.

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Simulation results GORIC(A) evidence synthesis

Simulation results GORIC(A) evidence synthesis (added approach) can be found in:

References:

Kuiper, R.M. and Clapper, E.-B. (2025). Aggregating Evidence for a Central Theory From Diverse Studies Using the Generalized Order-Restricted Information Criterion. Psychological Methods. Pre-print: https://osf.io/preprints/psyarxiv/qv76x_v2

Magnitude-hypotheses

Set of central theories regards height of effect size.

E.g., Cohen's d measured (in some studies), one could evaluate:

$$H_1 : d < 0,$$

$$H_2 : d > 0,$$

$$H_3 : d > 0.2,$$

$$H_4 : d > 0.5,$$

$$H_5 : d > 0.8.$$

$$H_1 : d < 0,$$

$$H_2 : 0 < d < 0.2,$$

$$H_3 : 0.2 < d < 0.5,$$

$$H_4 : 0.5 < d < 0.8,$$

$$H_5 : d > 0.8.$$

Now, overlapping hypotheses.

Now, range restrictions
(complexity as if equalities).

Or better: One of these versus its complement.

Or: incorporate such cut-off values when inspecting ordering of parameters (say, means; e.g., $\mu_1 - \mu_2 > 0.2 \times s.d.$).

Additional and alternative analyses

Other options for 'evSyn':

- *leave1studyout* function looks at outlier/influential studies:
Do evidence synthesis for all but one study.
- *priorICweights*
- *study_weights*

Future options:

- Variation measures
- Check publication bias via 'test' or (funnel) plot
- Subgroup analyses
- Account for dependency between studies (same sample or cohort)

Software

- R function *evSyn* in R package *restriktor*
- Interactive web application (Shiny app)
available from my site.